

TraceSetu problem and solution explained

A detailed guide for the team | 28 September 2026

An investigating officer has a cryptocurrency wallet address linked to a complaint. The officer needs to know which exchange or custodial service received the funds, and what evidence supports contacting that service. TraceSetu is the investigation workspace we have built to help answer that question.

1 What exactly is the problem

The official title supplied for our SIH challenge is “Automated Attribution of Unknown Cryptocurrency Wallets to Nearest Virtual Asset Service Providers (VASPs) through Blockchain Intelligence APIs”. The supplied context is the Ministry of Home Affairs, Indian Cyber Crime Coordination Centre (I4C), CIS Division, under Blockchain and Cybersecurity.

In cases involving fraud, ransomware, investment scams or other suspected crime, investigators may recover a wallet address from a complaint, payment record or device. A public blockchain can show recorded transfers involving that address. It usually does not show the customer name, account holder or the company responsible for every address.

The funds may move from the reported address to an intermediary, split across several addresses, change assets, or move to another blockchain before reaching an exchange. Looking through these movements manually can take considerable effort. A wrong ownership label or a missing branch can also send the investigator towards the wrong service.

The problem statement describes lawful requests being raised through SAHYOG. That workflow still needs a correctly identified recipient. If the receiving service is unknown, the officer may not know whom to ask for relevant customer or transaction records, or where to direct an authorised preservation or freezing request.

The question our system tries to answer

Starting from this reported wallet, which first receiving exchange or custodian can the available evidence support, through which transactions, and what missing evidence could change that answer?

“Nearest” means the fewest valid transfer steps to a supported receiving service within the examined data. It does not mean geographically nearest, cheapest, most popular, or necessarily where the money is now. A one-hop direct deposit and a two-hop route through an intermediary are different findings.

What solving it should achieve

An officer should receive a usable lead: the receiving service and address where supported, the transaction path, source records, limits of the search, and a reviewed next-action package. Identifying the beneficial owner and taking asset action require the appropriate records, authority and cooperation beyond the blockchain trail.

This guide explains the complete intended solution and the software already built. Pages 2–4 explain the concepts and example; pages 5–10 explain the workflow and technology; pages 11–12 cover verification, remaining work and team explanations.

2 The terms in everyday language

These terms describe different parts of an investigation. Keeping them separate prevents an address, a service and a person from being treated as the same thing.

Cryptocurrency and blockchain. A cryptocurrency is a digital asset, such as bitcoin. VDA means Virtual Digital Asset, the broader term used in the problem. A blockchain is a shared ledger that records transactions under a network's rules. Bitcoin, Ethereum, Tron, BNB Chain, Solana and Polygon are different networks. Some networks also support tokens: assets issued using software running on that network. A token name alone does not uniquely identify an asset.

Wallet and wallet address. A wallet is software or a service used to manage keys and interact with assets recorded on a blockchain. An address is an identifier used in those interactions, somewhat like a payment destination. One person may use many addresses. A service may manage addresses for many customers. The address itself is not a customer identity document.

VASP and custody. VASP means Virtual Asset Service Provider. In simple terms, it can be a business that exchanges, transfers or safeguards virtual assets for others. Our tracing task focuses on receiving exchanges and custodial services. Custody means a service controls assets or the means of access on a customer's behalf. In a self-custody or unhosted wallet, the user controls their own keys. An unlabelled address is not automatically self-custody. [S1]

Deposit wallet and hot wallet. A deposit address is used to receive customer deposits for a service. A hot wallet is an online operational wallet a service may use to move funds. A transfer into a wallet labelled "Exchange X" does not, by itself, prove it accepts customer deposits. We need evidence for the service identity and for the wallet's role.

Transaction hash and hop. A transaction hash is a transaction's identifying reference. A hop is one transfer step in a path. In $A \rightarrow B \rightarrow C$, reaching C takes two hops. When a transaction has several senders or recipients, the model must preserve that structure instead of pretending each apparent pair is a separate, proven payment.

API and blockchain intelligence. An application programming interface, or API, lets one program request data from another. A chain API may provide transaction history. An intelligence source may provide a claim that an address belongs to an exchange. Those are separate inputs. Blockchain intelligence combines observations and supporting assertions to produce an investigative explanation.

Bridge mixer and swap. A bridge or transfer protocol connects activity across networks. A swap exchanges one asset for another. DeFi means decentralised finance: financial activity implemented through blockchain programs called smart contracts. A mixer combines or obscures flows. These mechanisms can introduce ambiguity; similar amounts and times do not prove a connection.

Evidence provenance and confidence. Provenance means where a piece of evidence came from, when it was obtained and what it supports. A label is an assertion about an address, not an observed transfer. Confidence describes support for a conclusion; risk describes a reason for concern. TraceSetu currently uses explicit evidence grades and gaps rather than an invented probability of ownership or guilt.

Finality and reorganisation. A network's confirmation or finality rules describe when transactions can be treated as settled under its assumptions. A reorganisation changes which blocks form the accepted chain and can affect earlier observations. Our full design requires affected conclusions to be rechecked while preserving their earlier evidence.

3 Who uses it and what is required

The immediate users are investigating officers and technical analysts in Law Enforcement Agencies (LEAs). A reviewer checks consequential conclusions and request packages; an administrator manages access and operations. VASP compliance teams are intended external recipients of lawful requests. Victims benefit indirectly when an investigation can reach the right service sooner.

The built application has investigator, reviewer and admin roles, with agency and case boundaries. “Legal team” and “VASP compliance team” describe participants in the intended process; they are not extra implemented account types.

The problem statement broken into practical jobs

Requirement	What it means for the officer
Wallet intake and API integration	Accept a reported wallet, its blockchain, case reference and relevant dates. Ultimately receive authorised submissions from SAHYOG.
Nearest exchange or custodian	Find the first supported service on each path, including a direct deposit when the evidence supports that role.
Multiple blockchains	Handle Bitcoin, Ethereum, Tron, BNB Chain, Solana and Polygon with the correct rules for each network; extend through tested adapters.
Service and wallet classification	Distinguish sourced exchange clusters, deposit and hot wallets, custodians, mixers, bridges and swap services.
Tags and confidence	Show the source and strength of each identity assertion, contradictions and missing coverage.
Reports and reviewed routing	Give officers inspectable evidence and help prepare the correct disclosure, preservation or freezing-review package.
Dashboard and timely results	Organise cases, run searches, show progress, display fund movement and support investigation review.
Risk and optional intelligence	Explain sourced risk signals; support cross-chain mapping, laundering indicators and high-risk alerts where data and decoding permit.
Scale and operational reliability	Handle larger workloads with controlled costs, secure access, recoverable jobs, retained evidence and tested deployment.

What a useful result contains

The result should identify the case and search scope; show any supported service, receiving address and wallet role; list the chronological transfers and exact amounts; explain source quality, conflicts and missing branches; and suggest the next useful evidence request. A report must also distinguish a completed search from a complete view of reality.

A legitimate result can be “no supported VASP found in this acquired data”. It should explain why and what is missing. Public transaction records alone cannot supply every service label, customer record or legal contact.

4 One example from complaint to next action

This is an illustrative example to explain the logic. The wallets and Exchange Alpha below are invented; it is not a claim about a real case or a real exchange.

The reported wallet

An officer receives wallet A from a complaint and knows the relevant blockchain and dates. During that period, A sends 10 units of an asset to B. Later, B sends 8 units of the same asset to D. A source assertion identifies D as a customer deposit address operated by Exchange Alpha and is valid at the time of that arrival.

The supported path is $A \rightarrow B \rightarrow D$. D is two hops from A. If the transaction ordering and source assertion pass the checks, Alpha is a candidate receiving service. The report shows D as the receiving address and cites both the transfers and the separate deposit-role assertion.

Why the first service matters

Suppose D later sweeps funds into Alpha's hot wallet H. The first supported custody point is still D. Continuing through H and matching a later withdrawal to an unrelated customer would create a misleading trail. Once custody begins, the service's internal account records may be needed to answer the next question.

Why another branch can change the answer

A also sends funds to C, but C's history or ownership evidence is missing. C could be a nearer service at one hop. TraceSetu therefore cannot simply declare Alpha to be the nearest real VASP. It records the unresolved branch and explains that additional evidence at C could change the answer.

This is the distinction between "the nearest supported candidate in the data we examined" and "the nearest service anywhere in the real world". Even a fully fetched transaction history does not prove that every intermediary has been correctly identified.

What the officer can do next

The officer inspects the assertion on D and challenges its source. If that was the only support for Alpha, the candidate disappears from the challenged result. The original result remains available. This shows how much the answer depends on that source; it does not prove the source is false.

The next-query planner can prioritise missing history at C within a request budget. If the missing item is an ownership assertion, the next task may instead be manual evidence review or obtaining authorised intelligence. More transaction requests alone cannot manufacture an identity label.

What the amounts do and do not establish

The 10-unit and 8-unit transfers are observations. B may already have held other funds. A simple path does not prove that all 8 units came from the complaint, that one person controls A and B, or that Alpha still holds the assets. Current per-path exposure bounds can range from zero to the smallest relevant transfer amount. Overlapping paths cannot be added as if they were separate recovered funds.

After reviewing the evidence, an authorised officer can prepare a request addressed to a verified Alpha recipient. A separate reviewer approves its exact contents. The current software exports that package; it does not send it to SAHYOG or freeze anything.

5 How the system works behind the screen

Step 1 Define the case and search

The officer signs in, opens a case and supplies the blockchain, reported address, time window and search limits. Optional asset and amount filters narrow the scope. Address validation catches formatting errors; it does not verify ownership. An address-wide search is also different from proving how a particular victim's payment was allocated.

Step 2 Start a recorded job

The API checks permissions and creates a job. A background worker performs the longer task while the browser shows progress. The job records its inputs, status and attempts. Request, hop and graph limits prevent one investigation from expanding indefinitely. The same bound request can be recognised so a retry does not automatically create duplicate work.

Step 3 Acquire permitted data

A provider adapter translates our request into the selected service's API format. The worker collects available history, saves the original responses and records gaps, timeouts or missing access. Live mode never changes into a training example after a failure. Public or free access is the current constraint; account keys or provider permissions may still be required.

Step 4 Put different networks into a common model

Normalisation converts each provider response into a consistent record containing the network, transaction, sender or input set, recipient, asset, exact base-unit amount, time, ordering and source. Base units are the asset's smallest units, such as satoshis for bitcoin; keeping them as integers avoids rounding. Bitcoin requires links to specific spent outputs, called outpoints; address reuse alone is insufficient. [S2]

A snapshot freezes the acquired observations, assertions and coverage for a run. SHA-256 hashes act as file fingerprints. They help detect later changes to retained bytes; they do not independently prove that a provider supplied a truthful or complete account of the chain.

Step 5 Analyse chronological paths

The deterministic engine follows eligible successful transfers in valid order. It preserves separate branches and checks labels at the time funds arrived. Failed, pending, orphaned or out-of-scope events are excluded from confirmed paths. A custody boundary, mixer, unsupported transition, conflicting observation or exhausted search limit produces an explicit stop or unresolved branch.

Step 6 Explain and improve the answer

The engine returns candidates, paths, the graph, evidence grades, risk tags, unresolved frontiers and a nearestness certificate. Grades distinguish provider assertions, reviewed assertions and unverified hypotheses. Disagreeing supported labels remain conflicted. The officer can challenge a source, review missing evidence, or approve selected follow-up queries. A new analysis preserves the old one rather than silently rewriting case history.

The product-foundation branch adds versioned database upgrades, a separate worker process, worker heartbeat checks and an administrator health panel. These make the application easier to operate and diagnose. They do not turn a locally verified installation into a completed agency deployment.

6 The useful difference in our approach

Our proposed USP is an answer the officer can inspect, challenge and improve under a defined data budget. The contribution is the connected evidence workflow. Graphs, case dashboards and multiple blockchains are useful features, but they alone are not unique to TraceSetu.

First custody and an inspectable nearestness certificate

The first-custody frontier is the collection of earliest supported custodial endpoints across the searched branches. The certificate records the snapshot, search limits and gaps that affect the nearestness claim. It is a technical explanation, not a government certificate. Tested cases retain alternatives and prevent a stronger claim when an unresolved nearer branch could change the answer.

Challenge a source before relying on it

The officer can exclude a label, proof or source family and recompute the result. A source family groups related assertions so repeated claims from one origin are not mistaken for independent corroboration. The comparison shows candidates that disappear or remain, and branches that now need more evidence. The previous analysis is preserved.

Spend the next query where it can matter

The planner gives priority to unresolved nearer branches, then weighs observed path coverage against estimated request cost at the same depth. An officer reviews and executes selected queries. Manual reviews, unavailable providers and deferred queries remain visible. Estimated requests, reserved requests and successful responses are separate measurements; none automatically equals a provider's bill.

The execution workflow reserves request slots before dispatch, retains recoverable responses and prevents a consumed plan from being spent again with different inputs. Interrupted work does not reset its budget. Conflicting observations stay on hold rather than being silently selected. These controls have reproducible tests; percentage cost savings have not been measured.

Let another reviewer reproduce the result

An evidence package includes the snapshot, result, relevant source artifacts, report and a signed manifest. The verifier checks file membership, hashes and signature, then reruns the supported analysis version. Editing a file should fail verification. Reproducing a result establishes consistency with the retained inputs, not the truth of every ownership assertion.

Learn from reviewed service replies

A supported structured reply can become narrowly scoped evidence after checks for the request, signer, wallet, service and relevant period, followed by independent review. Corrections and withdrawals remain in the history and can trigger reassessment. This creates a controlled feedback process without turning an unverified email or a free-text comment into an ownership fact.

How we should demonstrate the value

Show a missing nearer branch, challenge a sole supporting label, execute a capped follow-up, detect a changed evidence file and reject a wrongly scoped service reply. To claim better real-world accuracy or efficiency, we still need independent cases and comparison under the same data and cost limits, measuring precision, coverage, abstention, investigator time and false routing.

7 What we have actually built

There are two published software baselines: the working SIH MVP on the private repository's main branch and additional verified foundation work on product/live-foundation. The latter was pushed as commit e834193. Development was stopped on 26 September; this document does not resume it.

An investigation workspace

The interface contains a case register, case membership, analysis setup and progress, a fund-movement graph, candidate and evidence inspection, gap explanations, source challenges, next-query planning and execution, reporting, requests, monitoring and audit views. A How it works guide is available before sign-in and within the application. Mobile layouts were checked for overflow. Job progress is visible, but data freshness depends on provider access and polling; there is no guaranteed continuous live feed.

Three explicitly different evidence modes

Training uses invented records to exercise the entire workflow. Import accepts structured snapshots or supported proof packages with validation and review controls. Live mode calls configured external providers and retains real responses. The result must show its mode; a complete training demonstration is not proof that the same service identities are available in public data.

Controlled permissions and review

Access depends on user role, agency or tenant, and case membership. A tenant is a separate organisation's access boundary in a shared application. The app uses password hashing, expiring sessions and CSRF protection, which checks authenticated browser changes against cross-site request forgery. Wrong-case and wrong-tenant access are tested.

Consequential recipient, evidence and request approvals require another authorised reviewer. Record versions detect conflicting updates. Training evidence cannot be used as if it were operational evidence. Passwords, session secrets, signing keys and private case data are excluded from the published source repository.

Correction and continued observation

The app supports withdrawal of reviewed assertions or proofs, identifies dependent analyses, blocks stale exports and creates superseding results. It also supports budgeted wallet watches and in-case change alerts while the worker is operating. An old downloaded report cannot be recalled automatically. Reliable external notification delivery and permissioned correction sharing across cases are still future work.

Operational foundation added after the MVP

Alembic migrations make database structure changes versioned and explicit. Existing databases are adopted only after schema checks; migration tests preserve case records. PostgreSQL tests cover concurrent worker claims and rollback of the heartbeat addition. A separate worker can run alongside the API, and saved heartbeats reveal when the worker becomes stale.

Readiness reports whether essential components are ready. Administrators see their own tenant's queue status and component checks. Operational logs use a small approved set of fields rather than copying case contents or provider URLs into application events. These are engineering controls already built; security certification and production operational governance remain separate requirements.

8 Blockchain coverage and difficult paths

An adapter being implemented means the code can interpret its supported response format. A successful public endpoint call means that endpoint responded. A validated transfer sample means actual transfer records were checked. None of these alone proves complete history or correct ownership attribution across that network.

EVM means Ethereum Virtual Machine, the execution system shared by several compatible networks. ERC20 and TRC20 describe token interfaces on Ethereum-style networks and Tron; SPL refers to Solana's program library and its token tooling. These need different decoders. A receipt records a transaction's execution outcome and events.

Network	Implemented behaviour	Recorded live evidence
Bitcoin	Esplora history and an explicitly selected BlockCypher alternative; input sets, outputs, exact satoshis and spent-output links.	A bounded BlockCypher sample acquired two transactions and three outputs; exact references and signed replay matched. No VASP identified.
Ethereum	Etherscan V2 adapter, normal/internal history indexes and supported ERC20 receipt checks.	Provider access was not configured in the latest check. No live transfer validation claimed.
BNB Chain	EVM adapter with its own network identity and the supported receipt checks.	Provider access was not configured; sharing EVM code does not validate this chain.
Polygon	EVM adapter; also one side of the scoped CCTP proof implementation.	Provider access was not configured; general live tracing remains unvalidated.
Tron	Confirmed native and TRC20 history with pagination and explicit gaps.	One public response, with no transfers in the bounded access probe. Not transfer validation.
Solana	Finalized signatures and supported parsed native/SPL transfers, including supported inner instructions.	One public response, with no transfers in the bounded access probe. Not transfer validation.

Cross chain support is deliberately specific

The built proof pipeline covers a bounded CCTP V2 native-USDC path between Ethereum and Polygon. USDC is a dollar-linked token. CCTP is Circle's Cross-Chain Transfer Protocol: it removes tokens on one network by burning them and creates corresponding tokens on another by minting them. Our implementation checks supported message bytes, signatures and destination receipt/mint details, including fee and arrival order. It has synthetic verification and review tests, not a completed real cross-chain acceptance case. [S3]

Other bridges, arbitrary swaps and unsupported contracts remain gaps. Imported proofs start as hypotheses until independently reviewed. A cryptographically consistent proof still depends on trusted key identity and source records; it does not identify the customer or receiving VASP by itself.

Clusters mixers risk and alerts

The evidence model can represent sourced exchange clusters and deposit/hot-wallet roles. Automatic ownership clustering is not implemented. A sourced mixer label stops deterministic continuation. Sourced risk tags can show exposure, but the app has no comprehensive ransomware, darknet, terrorism or fraud intelligence feed, no calibrated numerical risk score and no dedicated laundering-typology detector. Unknown remains unassessed, not safe.

9 Reports requests and service feedback

What leaves the analysis workspace

The officer can generate a readable PDF report, a CSV transfer table and structured JSON records. An evidence ZIP contains the snapshot and analysis, relevant retained raw artifacts, case metadata, an audit excerpt, metrics and a signed manifest. Reports carry the evidence mode, search scope, candidates, supporting paths, exact amounts, gaps and versions.

SHA-256 fingerprints detect altered bytes. Ed25519 is the digital-signature method used for the evidence manifest. A reviewer can verify the archive and reproduce the deterministic result with the supported engine version. Trust in the signer requires an independently obtained trusted key fingerprint. A valid signature does not establish an exchange identity or guarantee legal acceptance of the evidence.

The recipient must be verified separately

A wallet label may name a brand, whereas the legal recipient could be a specific entity in a particular jurisdiction. The app therefore maintains reviewed recipient records with entity, contact channel, verification source, expiry and purpose. No real service contacts are guessed or preloaded. Case relevance and the exact entity match are checked before preparing the request.

Three kinds of next action

A disclosure request seeks relevant records, which may include customer verification records, account activity or transaction references where available and lawfully requested. Customer verification is often called KYC, or Know Your Customer. A preservation request asks the recipient to retain specified information or evidence. A freezing-review package supports review of a proposed asset restriction. Selecting a type in software does not grant legal authority or establish which law applies.

The investigator supplies the basis and scope. A different authorised reviewer approves the exact request payload, meaning the actual recipient and contents. Changing that content invalidates the old approval. Evidence withdrawals, stale recipients and case permissions can prevent export. This reduces the chance of sending an outdated or misdirected package.

What the current SAHYOG boundary means

The software has its own authenticated APIs and exports. Official SAHYOG access, schema, credentials and delivery contract have not been supplied or integrated. Current request packages are explicitly NOT_SENT and NOT_CONNECTED. No real notices were sent and no assets were frozen. Official dispatch, delivery acknowledgements and service-action tracking belong to the planned integration.

How feedback becomes new evidence

The app can validate its supported structured response format against the original request ID and content hash, the reviewed recipient key, wallet and entity scope, time period and replay controls. A separate reviewer must approve promotion into a case assertion. A signed free-text statement does not automatically create a wallet ownership label.

A later correction can withdraw that assertion, flag affected results and require reassessment. The response format is TraceSetu's own contract, not an official VASP or SAHYOG standard. Real service identity onboarding, interoperable response formats and controlled reuse across investigations still require institutional arrangements.

10 The technology and the complete design

The graph is analysed by explicit rules in Python. There is no implemented large language model or machine-learning ownership predictor. The same supported evidence and analysis version should give the same result. This makes the reasoning inspectable and suitable for replay. CI means continuous integration: automated checks run when source changes are published.

Layer	Technology in the current code	Its job
User interface	React, TypeScript and Vite	Build the screens, forms and browser application.
Transfer map	Cytoscape.js	Draw and inspect the graph; it does not establish ownership.
Application API	Python, FastAPI and Pydantic	Validate inputs, enforce workflow rules and serve authenticated requests.
Case storage	SQLAlchemy with SQLite WAL; PostgreSQL tested in CI	Store users, cases, jobs, versions and audit records. WAL is SQLite's write-ahead log.
Schema upgrades	Alembic in the foundation branch	Apply versioned database changes with explicit legacy adoption.
Background work	Database jobs, leases and a separate worker option	Record, claim and recover bounded jobs; expose readiness and heartbeats.
Data acquisition	httpx and chain-specific adapters	Call permitted APIs and normalise supported observations.
Evidence and exports	Local content-addressed files, SHA-256, Ed25519 and ReportLab	Retain evidence, detect changes, sign manifests and create PDFs.
Quality checks	pytest, Ruff, TypeScript, Playwright and GitHub Actions	Check backend behaviour, code consistency, builds and browser workflows.

The full product we intend to deliver

The complete service would accept authorised case intake from SAHYOG or another case system, check available data coverage, run recoverable multi-chain acquisition, resolve sourced service assertions, analyse first custody, and give officers the graph, evidence, uncertainty and next-query plan. Reviewed reports and requests would then pass through an authorised delivery adapter, with acknowledgements, replies, corrections and monitoring tied back to the case.

What larger deployment still requires

The research proposes agency single sign-on and multi-factor authentication, governed recipient and asset registries, encrypted evidence storage with retention and legal holds, controlled signing keys, shared provider quotas, fair job scheduling, recovery drills and operational monitoring. Temporal was proposed for durable workflows and a Neo4j projection only if scale measurements justify it. Neither is running in the delivered application.

PostgreSQL and migration tests are now real progress beyond the earlier prototype plan. They do not establish a deployed agency service. Load tests, backup restoration, end-to-end chain reorganisation handling, independent security assessment and operating ownership remain necessary before a production acceptance claim.

11 What the verification establishes

Working software and recorded checks

The preserved MVP was published to a private GitHub repository. The foundation branch adds the operational and validation work described here. The final recorded commit is e834193; its backend, frontend and PostgreSQL GitHub checks passed on 26 September 2026. These are test results for that code, not an ownership accuracy percentage. [P1]

The local full suite recorded 128 passing tests and two PostgreSQL-only skips. The hosted PostgreSQL suite recorded 130 passing tests, including permission boundaries, migration preservation and concurrent claims. Browser checks covered login, cases, analysis, query planning, source challenge, bundle verification, coverage and mobile layout. The separate API/worker workflow and admin health panel also passed.

The actual public Bitcoin check

A bounded live acquisition examined the public Bitcoin address 13AM4VW2dhxYgXeQepoHkHSQuy6NgaEb94 for 3 August 2017 UTC. With BlockCypher explicitly selected, the application acquired two transactions and three outputs in one successful provider response. Three exact reference matches and signed evidence replay passed. No receiving VASP identity was established.

The separately fetched reference used the same provider. The check supports correct transport, parsing, amount comparison and replay for that sample. It is not independent attribution ground truth. “Live” here means an actual API fetch of real historical records, not a claim that the transaction happened during the demonstration. Broader transfer coverage on the other five chains remains unvalidated.

What is still unfinished

The full-product ledger records 52 of 95 named tasks verified, with 43 remaining: 31 engineering or validation tasks and 12 external acceptance gates. These are unweighted task counts, not 55 percent production readiness. The scoped MVP checklist and the full-product checklist measure different things.

Engineering work still includes broader chain and asset semantics, robust canonical-chain and reorganisation reconciliation, supported additional bridges/swaps, governed clustering and typology rules, operational notifications, shared quotas and fairness, full deployment, recovery and security checks. Typologies are recognisable patterns, such as rapid dispersal or repeated layering of transfers; a pattern alone does not prove crime. The container deployment was not verified locally.

External requirements include lawful data entitlements and reliable ownership labels, independently verified test cases, trusted service identities and recipients, official SAHYOG contracts, agency access and retention policies, and acceptance by the responsible users and institutions. Public/free APIs can provide valuable chain facts but do not remove these dependencies.

How success must eventually be measured

Measure correct first-service answers together with answer coverage, abstentions, false recipients, cost, latency and investigator time. Use independent ground truth, including difficult and unresolved cases. A vendor’s own label cannot serve as both the predicted answer and independent proof that the answer was correct. Asset recovery or freezing outcomes require separate evidence of actual authorised action.

12 Questions every team member should answer

Why can an officer not just use a blockchain explorer

An explorer shows useful records for one network. Our application organises a case, preserves source evidence, searches supported paths, separates service claims from transfers, explains gaps, challenges sources, controls follow-up requests and prepares reviewed exports. An explorer can still be one of the investigation's supporting tools.

How do we know which company owns an address

We need a sourced assertion valid for that wallet and time, or a suitably authenticated and reviewed service confirmation. A graph pattern can suggest a hypothesis. It cannot by itself supply customer identity, historical ownership or proof that the address accepts deposits.

Does finding an exchange mean we found the criminal

No. It identifies a possible service that may hold relevant records. The customer relationship, beneficial owner, account activity and connection to the complaint need separate evidence and lawful investigation. A receiving exchange is not automatically responsible for the suspected offence.

Does the prototype already do the complete solution

It demonstrates a substantial end-to-end independent workflow, with real bounded Bitcoin acquisition and tested training/imported evidence paths. Six adapters exist, but broad live attribution, general cross-chain coverage, calibrated confidence and official SAHYOG action are not complete. The right description is a working MVP plus a verified product foundation.

How should we explain TraceSetu in a meeting

An officer may have a crypto wallet address but no idea which service to contact. TraceSetu follows the available transfer records to the first supported exchange or custodian. It shows the evidence, explains what is missing and lets the officer test whether the answer depends on a disputed source. It can then prepare a reviewed evidence and request package. Our goal is to help officers make a defensible next move; the service or authorised institution remains responsible for records disclosure and asset action.

Sources and where to read further

S1 FATF guidance on virtual assets and VASPs. Background terminology, not a statement of case-specific legal authority.
<https://www.fatf-gafi.org/en/publications/Fatfrecommendations/Guidance-rba-virtual-assets-2021.html>

S2 Bitcoin developer guide on transactions and spent-output references.
<https://developer.bitcoin.org/devguide/transactions.html>

S3 Circle documentation on CCTP. Our supported subset is described on page 8.
<https://developers.circle.com/cctp>

P1 Published foundation source and final recorded automated checks. Repository access is private.
<https://github.com/notvinayaks/TraceSetu/tree/product/live-foundation>
<https://github.com/notvinayaks/TraceSetu/actions/runs/36226593776>

Project records: skills/custody-atlas-sih/SKILL.md contains the complete handover and supplied problem statement; research/report.md and research/sources.json contain the original research and 44-source register; docs/PRODUCT_FOUNDATION_VERIFICATION.md records the foundation checks; IMPLEMENTATION_CHECKLIST.md lists the unfinished full-product work.